Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
NOVEMBER/DECEMBER 2024
SUBJECT: CIPHER SYSTEM (ECE 4069), PE-V

TIME: 3 HOURS **MAX. MARKS: 50**

Instructions to candidates
- Answer **ALL** questions.
- Missing data may be suitably assumed.

Q. No.	Questions	M*	C*	A*	B*
1A.	Determine the multiplicative inverse of 314 mod 841.	3	1	1,2,3,6,9,10, 12	3
1B.	Construct the GF (2^3) using primitive polynomial $p(x)=1+x+x^3$. Determine Inverse of $(1+\alpha^3+\alpha^5)$ for the same.	3	1	1,2,3,6,9,10, 12	4
1C.	Solve using repeated squaring technique 215^173 mod841.	4	1	1,2,3,6,9,10, 12	3
2A.	Determine the plain text from the received cipher text "**.FJLXJ**" using Hill cipher with encryption key A $\begin{vmatrix} 3 & 7 & 1 \\ 5 & 9 & 11 \\ 1 & 19 & 13 \end{vmatrix}$ Assume that the alphabet A=-Z have numerical values 0-25 respectively, & 'dot'=26.	5	2	1,2,3,6,8,9,10,12	4
2B.	Determine the plain text for the received cipher text "HKOIPHITVM" using Playfair Cryptosystem with Key: EXAM.	3	2	1,2,3,6,8,9,10,12	3
2C.	In affine block cipher with encryption key A & T, plain text m is encrypted using the rule f(m)=Am+T, then Show that f(summation(mi)) =summation f(m1)+f(m2)+1 $f(\sum_{i=1}^{n}(m_i)) = \sum_{i=1}^{n} f(m_i) + \mathrm{t}\ \partial_n;$ where $\partial_n = 1\ if\ n = even$ and $\partial_n = 0\ otherwise$	2	3	1,2,3,6,8,9,12	
3A.	Determine the cipher text using S-DES cryptosystem on plaintext 00101000, using s-DES with encryption key: 1100011110.	5	3	1,2,3,6,8,9,12	
3B.	Calculate the Sub Byte transformation of 0xEB. Show the computation steps clearly. Verify the result using the table.	3	3	1,2,3,6,8,9,12	4
3C.	Explain the type of Weak keys in DES algorithm.	2	3	1,2,3,6,8,9,12	3

4A.	Bob uses ElGamal cryptosystem with prime number p=113, primitive root g=6 and secret key d =23. Determine the Bobs public key. Determine the cipher text that Alice encrypted the plaintext "HIDE" with private key r=9 and sent to Bob. Consider 26-letter alphabet, A-Z with numerical values 0-25 respectively, with plain text represented as monogram and cipher text bigram.	5	4	1,2,3,4,6,8,9,10,12	
4B.	Merkle Hellman Knapsack cryptosystem uses private key (40, 213) and sequence { 2,5,15,40,65}. Determine the public keys. Encrypt the plain text ATTACK. Use the 26 letter alphabet 26-letter alphabet, A-Z with numerical values 0-25 respectively to represent the plain text.	3	4	1,2,3,4,6,8,9,10,12	
4C.	Determine the inverse of $2(g^2, g^6)$ for Elliptic curve defined by $E_{2\wedge3}(g^3,1)$ with $p(x)= x^3 + x+1$.	2	4	1,2,3,4,6,8,9,10,12	
5A.	An elliptic curve is defined by $E_{97}(3,21)$. Determine the 3P if point P is (2,36).	5	5	1,2,3,4,6,8,9,10,12	
5B.	Show the computation of values of constants in message digest initialization in SHA512.	3	5	1,2,3,4,6,8,9,10,12	
5C.	Explain the following terms 1. Confusion - Diffusion 2. Message Integrity	2	5	1,2,3,4,6,8,9,10,12	

M*--Marks, C*--CLO, A*--AHEP LO, B* Blooms Taxonomy Level